

Insu Yun

Associate Professor (Untenured)
School of Electrical Engineering,
Korea Advanced Institute of Science and Technology (KAIST)

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Research Interests

System security, software security, binary analysis, fuzzing, and applied cryptography.

Education

Georgia Institute of Technology Ph.D. in Computer Science Advisor: Dr. Taesoo Kim	Aug. 2015 – Dec. 2020
Korea Advanced Institute of Science and Technology (KAIST) B.S. in Computer Science & Mathematics	Sep. 2008 – Feb. 2015

Work Experience

KAIST , Daejeon, South Korea Associate Professor Assistant Professor	Feb. 2021 – Mar. 2025 – Present Feb. 2021 – Feb. 2025
Microsoft Research , Research Intern, Seattle, WA Contributed to REPT, a system that utilizes Intel Processor Trace to diagnose production failures Mentor: Weidong Cui	May. 2017 – Aug. 2017
Georgia Tech , Research Assistant, Atlanta, GA	Aug. 2015 – Dec. 2020
Korean Cyber Command , Software Developer, Seoul, Korea Served for the mandatory military service	Apr. 2012 – Jan. 2014

Publications

* Students advised by myself are underlined.

International Conferences (**Top-tier** and others)

30. **PatchIsland: Orchestration of LLM Agents for Continuous Vulnerability Repair (to appear)**
Wonyoung Kim*, Seunggi Min*, Minjae Gwon, Haein Lee, Dowoo Baik, Hyeon Heo, Minjae Lee, Min Woo Baek, Yonghui Jin, Younggi Park, Yunjae Choi, Taesoo Kim, Sangdon Park, and **Insu Yun**
Proceedings of the 42nd Annual Computer Security Applications Conference (ACSAC 2026)
Los Angeles, CA, December 2026
29. **MTEscape: Bypassing Asynchronous Kernel MTE via Conventional Memory Corruption Vulnerabilities (to appear)**
Kyeongmin Kim, and **Insu Yun**
Proceedings of the 33rd ACM Conference on Computer and Communications Security (**CCS 2026**)
The Hague, The Netherlands, November 2026
28. **Prism: A Multi-Team Orchestration of LLM Agents for Automatic Program Repair (to appear)**
Yunjae Choi, Min Woo Baek, and **Insu Yun**

27. **QueryHouse: Cross-DBMS Differential Testing with LLM and Query Transpilation (to appear)**
Seoyeon Oh, Juhyun Song, Jaemin Park, Kwanghee Lee, Minae Hyeon, Minji Kim, Sehyuk Ka, Gwangwun Jung, Brian Pak, and **Insu Yun**
Proceedings of the 31st European Symposium on Research in Computer Security (ESORICS 2026)
Rome, Italy, September 2026
Best Paper Runner-up Award (3 out of 882 submissions)
26. **RTCon: Context-Adaptive Function-Level Fuzzing for RTOS Kernels**
Eunkyu Lee, Junyoung Park, and **Insu Yun**
Proceedings of the 2026 Annual Network and Distributed System Security Symposium (NDSS 2026)
February 2026
25. **OTABase: Enhancing Over-the-Air Testing to Detect Memory Crashes in Cellular Basebands**
CheolJun Park, Marc Egli, BeomSeok Oh, Tuan Dinh Hoang, Suhwan Jeong, Martin Crettol, **Insu Yun**, Mathias Payer, and Yongdae Kim
Proceedings of the 41st Annual Computer Security Applications Conference (ACSAC 2025)
Austin, TX, December 2025
24. **CROSS-X: Generalized and Stable Cross-Cache Attack on the Linux Kernel**
Dongok Kim*, Juhyun Song*, and **Insu Yun**
Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS 2025)
Taipei, Taiwan, October 2025
(Acceptance rates: 14%, 316/2278)
23. **Windows plays Jenga: Uncovering Design Weaknesses in Windows File System Security**
Dong-uk Kim*, Junyoung Park*, Sanghak Oh, Hyoungshick Kim, and **Insu Yun**
Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS 2025)
Taipei, Taiwan, October 2025
(Acceptance rates: 14%, 316/2278)
22. **LLFuzz: An Over-the-Air Dynamic Testing Framework for Cellular Baseband Lower Layers**
Tuan Dinh Hoang, Taekkyung Oh, CheolJun Park, **Insu Yun**, and Yongdae Kim
Proceedings of the 34th USENIX Security Symposium (Security 2025)
Seattle, WA, August 2025
(Acceptance rates: 17%, 407/2385)
21. **Too Much of a Good Thing: (In-)Security of Mandatory Security Software for Financial Services in South Korea**
Taisic Yun, Suhwan Jeong, Yonghwa Lee, Seungjoo Kim, Hyoungshick Kim, **Insu Yun**, and Yongdae Kim
Proceedings of the 34th USENIX Security Symposium (Security 2025)
Seattle, WA, August 2025
(Acceptance rates: 17%, 407/2385)
20. **FirmState: Bringing Cellular Protocol States to Shannon Baseband Emulation**
Suhwan Jeong, Beomseok Oh, Kwangmin Kim, **Insu Yun**, Yongdae Kim, and CheolJun Park
Proceedings of the 18th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec 2025)
Arlington, VA, June 2025
19. **Automated Attack Synthesis for Constant Product Market Makers**
Sujin Han, Jinseo Kim, Sung-Ju Lee, and **Insu Yun**

Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis 2025 ([ISSTA 2025](#))
Trondheim, Norway, June 2025
(Acceptance rates: 19%, 107/550)

18. **Bridging the Gap between Real-World and Formal Binary Lifting through Filtered-Simulation**
Jihee Park, **Insu Yun**, and Sukyoung Ryu
Proceedings of the ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications 2025 ([OOPSLA 2025](#))
Singapore, October 2025
(Acceptance rates: 44%, 67/151)
17. **RGFuzz: Rule-Guided Fuzzer for WebAssembly Runtimes**
Junyoung Park, Yunho Kim, and **Insu Yun**
Proceedings of the 46th IEEE Symposium on Security and Privacy ([Oakland 2025](#))
San Francisco, CA, May 2025
(Acceptance rates: 15%, 256/1739)
16. **BaseComp: A Comparative Analysis for Integrity Protection in Cellular Baseband Software**
Eunsoo Kim*, Min Woo Baek*, CheolJun Park, Dongkwan Kim, Yongdae Kim, and **Insu Yun**
Proceedings of the 32nd USENIX Security Symposium ([Security 2023](#))
Anaheim, CA, August 2023
(Acceptance rates: 31%, 442/1444)
15. **QueryX: Symbolic Query on Decompiled Code for Finding Bugs in COTS Binaries**
HyungSeok Han, JeongOh Kyea, Yonghwi Jin, Jinoh Kang, Brian Park, and **Insu Yun**
Proceedings of the 44th IEEE Symposium on Security and Privacy ([Oakland 2023](#))
San Francisco, CA, May 2023
(Acceptance rates: 17%, 196/1147)
14. **Fuzzing@Home: Distributed Fuzzing on Untrusted Heterogeneous Clients**
Daehee Jang, Ammar Askar, **Insu Yun**, Stephen Tong, Yiqin Cai, and Taesoo Kim
Proceedings of the 2022 International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2022)
October 2022
13. **DoLTest: In-depth Downlink Negative Testing Framework for LTE Devices**
CheolJun Park*, Sangwook Bae*, BeomSeok Oh, Jiho Lee, Eunkyu Lee, **Insu Yun**, and Yongdae Kim
Proceedings of the 31th USENIX Security Symposium ([Security 2022](#))
Boston, MA, August 2022
(Acceptance rates: 18%, 256/1414)
12. **HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators**
Insu Yun, Woosun Song, Seunggi Min, and Taesoo Kim
Proceedings of the 28th ACM Conference on Computer and Communications Security ([CCS 2021](#))
Seoul, South Korea, November 2021
(Acceptance rates: 22%, 196/880)
11. **Preventing Use-After-Free Attacks with Fast Forward Allocation**
Brian Wickman, Hong Hu, **Insu Yun**, Daehee Jang, JungWon Lim, Sanidhya Kashyap, and Taesoo Kim
Proceedings of the 30th USENIX Security Symposium ([Security 2021](#))
Vancouver, B.C., Canada, August 2021
(Acceptance rates: 19%, 246/1316)
10. **BaseSpec: Comparative Analysis of Baseband Software and Cellular Specifications for L3 Protocols**

Eunsoo Kim*, Dongkwan Kim*, Cheoljun Park, **Insu Yun**, and Yongdae Kim

Proceedings of the 2021 Annual Network and Distributed System Security Symposium (**NDSS 2021**)

February 2021

(Acceptance rates: 15%, 87/578)

9. **Automatic Techniques to Systematically Discover New Heap Exploitation Primitives**

Insu Yun, Dhaval Kapil, and Taesoo Kim

Proceedings of the 29th USENIX Security Symposium (**Security 2020**)

Boston, MA, August 2020

(Acceptance rates: 16%, 157/977)

8. **Fuzzing JavaScript Engines with Aspect-preserving Mutation**

Soyeon Park, Wen Xu, **Insu Yun**, Daehee Jang, and Taesoo Kim

Proceedings of the 41st IEEE Symposium on Security and Privacy (**Oakland 2020**)

San Francisco, CA, May 2020

(Acceptance rates: 12%, 104/841)

Nominated as a finalist in CSAW Best Applied Research Paper Award 2020

7. **REPT: Reverse Debugging of Failures in Deployed Software**

Weidong Cui, Xinyang Ge, Baris Kasikci, Ben Niu, Upamanyu Sharma, Ruoyu Wang, and **Insu Yun** (alphabetical)

Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (**OSDI 2018**)

Carlsbad, CA, October 2018

(Acceptance rates: 18%, 47/257)

Jay Lepreau Best Paper Award (3 out of 257 submissions)

6. **QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing**

Insu Yun, Sangho Lee, Meng Xu, Yeongjin Jang, and Taesoo Kim

Proceedings of the 27th USENIX Security Symposium (**Security 2018**)

Baltimore, MD, August 2018

(Acceptance rates: 19%, 100/524)

Distinguished Paper Award (5 out of 524 submissions)

5. **CAB-Fuzz: Practical Concolic Testing Techniques for COTS Operating Systems**

Su Yong Kim, Sangho Lee, **Insu Yun**, Wen Xu, Byoungyoung Lee, Youngtae Yun, and Taesoo Kim

Proceedings of the 2017 USENIX Annual Technical Conference (ATC 2017)

Santa Clara, CA, July 2017

4. **APISan: Sanitizing API Usages through Semantic Cross-checking**

Insu Yun, Changwoo Min, Xujie Si, Yeongjin Jang, Taesoo Kim, and Mayur Naik

Proceedings of the 25th USENIX Security Symposium (**Security 2016**)

Austin, TX, August 2016

(Acceptance rates: 16%, 72/463)

Nominated as a finalist in CSAW Best Applied Research Paper Award 2016

3. **HDFI: Hardware-Assisted Data-Flow Isolation**

Chengyu Song, Hyungon Moon, Monjur Alam, **Insu Yun**, Byoungyoung Lee, Taesoo Kim, Wenke Lee, and Yunheung Paek

Proceedings of the 37th IEEE Symposium on Security and Privacy (**Oakland 2016**)

San Jose, CA, May 2016

(Acceptance rates: 13%, 55/413)

2. **Analyzing Security of Korean USIM-based PKI Certificate Service**

Shinjo Park, Suwan Park, **Insu Yun**, Dongkwan Kim, and Yongdae Kim

Proceedings of the 15th International Workshop on Information Security Applications (WISA 2014)

Jeju Island, Korea, August 2014

1. **Kargus: A Highly-scalable Software-based Intrusion Detection System**

Muhammad Jamshed, Jihyung Lee, Sangwoo Moon, **Insu Yun**, Deokjin Kim, Sungryoul Lee, Yung Yi, and KyoungSoo Park

Proceedings of the 19th ACM Conference on Computer and Communications Security (**CCS 2012**)

Raleigh, NC, October 2012

(Acceptance rates: 19%, 81/426)

International Journal

1. **Scalable and Secure Virtualization of HSM with ScaleTrust**

Juhyeng Han, **Insu Yun**, Seongmin Kim, Taesoo Kim, Soeul Son, and Dongsu Han

IEEE/ACM Transactions on Networking (ToN)

November 2022

Other Refereed Materials

5. **CTFusion: A CTF-based Benchmark for LLM Agent Evaluation**

Dongjun Lee, Ga-eun Bae, and **Insu Yun**

Proceedings of the Second Workshop on Agents in the Wild: Safety, Security, and Beyond (ICML AIWILD 2026)

July 2026

4. **From the Vulnerability to the Victory: A Chrome Renderer 1-Day Exploit's Journey to v8CTF Glory**

Haein Lee, and **Insu Yun**

TyphoonCon 2024

Seoul, Korea, May 2024

3. **One shot, Triple kill: Pwning all three Google kernelCTF instances with a single 1-day Linux vulnerability**

Dongok Kim, Seunghyun Lee, and **Insu Yun**

POC 2023

Seoul, Korea, November 2023

2. **Compromising the macOS kernel through Safari by chaining six vulnerabilities**

Yonghwi Jin, Jungwon Lim, **Insu Yun**, and Taesoo Kim

Black Hat USA Briefings (Black Hat USA 2020)

Las Vegas, NV, August 2020

1. **AVPASS: Leaking and Bypassing Antivirus Detection Model Automatically**

Jinho Jung, Chanil Jeon, Max Wolotsky, **Insu Yun**, and Taesoo Kim

Black Hat USA Briefings (Black Hat USA 2017)

Las Vegas, NV, July 2017

Thesis

1. **Concolic Execution Tailored for Hybrid Fuzzing**

Insu Yun

Ph.D. thesis, Georgia Institute of Technology

Atlanta, GA, December 2020

Professional Activities

Technical Program Committee (International)

Program Committee, *IEEE Symposium on Security and Privacy (Oakland)*, 2024–2026

Program Committee, *Network and Distributed System Security Symposium (NDSS)*, 2024–2026

Program Committee, *The International Conference on Cryptology and Network Security (CANS)*, 2025

Program Committee, *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2022–2023

Journal Editor (International)

Associate Editor, *ACM Transaction on Storage (ToS)*, 2024–2026

Others (International)

Organization Committee (Publicity Chair), *Network and Distributed System Security Symposium (NDSS)*, 2027

Organization Committee (Web Chair), *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2024

Artifact Evaluation Committee, *ACM Conference on Computer and Communications Security (CCS)*, 2023

Artifact Evaluation Committee, *USENIX Security Symposium (Security)*, 2023

Organization Committee (Web Chair), *ACM Conference on Computer and Communications Security (CCS)*, 2021

Technical Program Committee (Domestic)

Program Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2026

Program Committee, *Conference on Information Security and Cryptography Winter (CISC-W)*, 2025

Program Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2021

Others (Domestic)

Policy Advisory Committee, *Ministry of National Defense*, 2026

Security Special Committee, *National Artificial Intelligence Strategy Committee*, 2026

Security Task Force, *National Artificial Intelligence Strategy Committee*, 2025

Advisory Boards, *AI Cyber Defense Contest (ACDC)*, 2025

Advisory Boards, *HackTheon Sejong*, 2024–2025

Organization Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2021

Teaching Experience

Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST)	Fall 2025
• Evaluation – Average: 4.29 / 5	
Software Hacking Theory and Practice (EE517 at KAIST)	Spring 2025
• Evaluation – Average: 4.67 / 5	
Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST)	Fall 2024
• Evaluation – Average: 4.54 / 5	
Information Security Laboratory (IS521 at KAIST)	Fall 2024
• Evaluation – Average: 4.64 / 5	
Software Hacking Theory and Practice (EE517 at KAIST)	Spring 2024
• Evaluation – Average: 4.92 / 5	
Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST)	Fall 2023
• Evaluation – Average: 4.57 / 5	
Software Hacking Theory and Practice (EE517 at KAIST)	Spring 2023
• Evaluation – Average: 4.54 / 5	
Programming Structures for Electronical Engineering (EE209 at KAIST)	Fall 2022
• Evaluation – Average: 4.65 / 5	
Software development environment and tools practice (EE485-A at KAIST)	Fall 2022
• Evaluation – Average: 4.43 / 5	
Software Security (EE595-B at KAIST)	Spring 2022
• Evaluation – Average: 5 / 5	
Programming Structures for Electronical Engineering (EE209 at KAIST)	Fall 2021
• Evaluation – Average: 4.34 / 5	

Software development environment and tools practice (EE485-A at KAIST)

Fall 2021

- Evaluation – Average: 4.34 / 5

Software Security (EE595-B at KAIST)

Spring 2021

- Evaluation – Average: 4.9 / 5

Honors & Awards

Academic awards

Best Paper Runner-up Award, ESORICS 2026 (3 out of 882 submissions)	Sep. 2026
KAIST Q-Day Special Commendation for Research, KAIST	Nov. 2025
Song-Am Future Researcher Award, KAIST	Feb. 2025
Technology Innovation Awards, KAIST College of Engineering	Dec. 2024
Best Teaching Award, KAIST Electrical Engineering	Oct. 2024
Prize for Excellence in Teaching, KAIST	Feb. 2024
Frontiers of Science Award, The First International Congress of Basic Science (ICBS)	July. 2023
Best Teaching Award, KAIST Electrical Engineering	Sep. 2021
Jay Lepreau Best Paper Award, USENIX OSDI 2018	Aug. 2018
Distinguished Paper Award, USENIX Security 2018	Aug. 2018

Hacking competitions

DEFCON 34 CTF, 2nd place (Team 0x4b52)	Aug. 2026
DEFCON 26 CTF, 1st place (Team DEFKOR00T)	Aug. 2018
DEFCON 24 CTF, 3rd place (Team DEFKOR)	Aug. 2016
DEFCON 23 CTF, 1st place (Team DEFKOR)	Aug. 2015
Whitehat contest 2014, 1st place (Team SysSec)	Nov. 2014
DEFCON 22 CTF, 10th place (Team GoN)	Aug. 2014
SECCON CTF 2014, 1st place (TOEFL Beginner)	Feb. 2014
Codegate CTF 2012, 3rd place (Team GoN)	Apr. 2012
Secuinside CTF, 3rd place (Team GoN)	Oct. 2011
ISEC CTF, 1st place (Team GoN)	Sep. 2011
DEFCON 18 CTF, 3rd place (Team GoN)	Aug. 2010
Codegate CTF 2010, 5th place (Team GoN)	Apr. 2010
KISA HDCON, Gold Medal, 2nd place (Team GoN)	May 2009
Codegate CTF 2009, 4th place (Team GoN)	Apr. 2009

Scholarships

National Research Foundation of Korea Scholarship for Undergraduate	Mar. 2008 – Dec. 2013
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Others

DARPA AIxCC, 1st place (Team Atlanta), \$4M award	Aug. 2025
Cyber Security Challenge, 2nd place (Team HackingLab), \$400K research grant	2023
DARPA Cyber Grand Challenge, Finalist (Team Disekt)	Aug. 2016
KISA Bug Bounty Program's Hall of Fame	2013

Vulnerability Discovery Reward (aka Bug bounty)

To summarize, \$244.6K (by my students) and \$92.8K (by myself) bug bounties are awarded so far.

By my students

Pwn2Own - Microsoft Edge and Google Chrome (\$145K), ZDI, by SeunHyun Lee	Mar. 2024
v8CTF - CVE-2023-6702 (\$10K), Google, by Haein Lee	Jan. 2024

kernelCTF - CVE-2023-3390 (\$67.8K), Google, by Dongok Kim and SeunHyun Lee	Oct. 2023
Type confusion in V8 (\$7K), Google, by Haein Lee	Mar. 2023
NAS authentication bypass in Exynos (\$14.8K), Samsung Electronics, by Eunsoo Kim and CheolJun Park	Feb. 2022

By myself

PSV-2021-0304: afpd auth bypass (\$300), NETGEAR	Mar. 2021
Pwn2Own Apple Safari with a kernel privilege escalation (\$70K), ZDI, with Yonghwi Jin and Jungwon Lim	Mar. 2020
Apple Safari sandbox escape (\$20K), Apple	Dec. 2019
Three integer overflow vulnerabilities in PHP (\$1.5K), the Internet Bug Bounty	Jun. 2016
An Integer Overflow in Python zipimport (\$1K), the Internet Bug Bounty	Apr. 2016

Patents

International

3. Dynamic testing method and system for detecting memory corruptions in lower layers of cellular baseband of mobile communication devices (Pending)

Inventors: Yongdae Kim, Hoang Dinh Tuan, Insu Yun, Taekkyung Oh, Beomseok Oh, Mincheol Son, CheolJun Park

Application date: 2026.01.07

Application number: 19442669

Country: US

2. Security analysis system and method based on negative testing for protocol implementation of LTE device

Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun

Registration date: 12511402

Patent number: 2025.12.30

Country: US

1. Reverse debugging of software failures

Inventors: Weidong Cui, Xinyang Ge, Baris Kasikci, Cengiz Can, Ben Niu, Ruoyu Wang, Insu Yun

Registration date: 10565511

Patent number: 2020.02.18

Country: US

Domestic

9. Method and apparatus for detecting memory vulnerabilities based on network-side state control in wireless communication systems (Pending)

Inventors: Yongdae Kim, Beomseok Oh, Insu Yun, Mincheol Son, Hoang Dinh Tuan, Soohwan Jeong, CheolJun Park

Application date: 2025.12.03

Application number: 10-2025-0189113

Country: Korea

8. Method and apparatus for fuzzing-based vulnerability detection (Pending)

Inventors: Insu Yun, Eunkyu Lee, Junyeong Park

Application date: 2025.11.11

Application number: 10-2025-0169729

Country: Korea

7. Dynamic testing method and system for detecting memory vulnerabilities in lower layers of cellular baseband of mobile communication devices (Pending)

Inventors: Yongdae Kim, Taekkyung Oh, Beomseok Oh, Mincheol Son, CheolJun Park, Hoang Dinh Tuan, Insu Yun

Application date: 2025.05.07

Application number: 10-2025-0059118

Country: Korea

6. Heuristic-based binary lifting and verification system (Pending)

Inventors: Sukyoung Ryu, Jihee Park, Insu Yun

Application date: 2025.03.31

Application number: 10-2025-0041131

Country: Korea

5. Method and apparatus for decompilation-specialized static binary translation

Inventors: Insu Yun, Woosun Song, Seunggi Min

Registration date: 10-2834292-0000

Patent number: 2025.07.10

Country: Korea

4. Security analysis system and method based on negative testing for protocol implementation of LTE device

Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun

Registration date: 10-2514797-0000

Patent number: 2023.03.23

Country: Korea

3. Method and system for automatically analyzing bugs in cellular baseband software using comparative analysis based on cellular specifications

Inventors: Yongdae Kim, Eunsoo Kim, Dongkwan Kim, CheolJun Park, Insu Yun

Registration date: 10-2546946-0000

Patent number: 2023.06.20

Country: Korea

2. Methods and systems for key management service provision

Inventors: Dongsoo Han, JuHyeng Han, Insu Yun

Registration date: 10-2803138-0000

Patent number: 2025.04.28

Country: Korea

1. Security analysis system based on negative testing for protocol implementation of LTE device (Pending)

Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun

Application date: 2021.10.07

Application number: 10-2021-0133149

Country: Korea

Invited Talks

International

- Title: Atlantis: an Autonomous LLM-Powered Bug Finding and Fixing System
– Presented at KAIST-NYU Workshop Nov. 2025
- Title: How to build Skynet — a system that hacks systems
– **Keynote** speech at TyphoonCon, Seoul, Korea Jun. 2023
- Title: HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators
– Presented at ACM CCS 2021, Online Nov. 2021
- Title: Automatic Techniques to Systematically Discover New Heap Exploitation Primitives
– Presented at USENIX Security 2020, Online Aug. 2020
- Title: QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing
– Presented at USENIX Security 2018, Baltimore, MD Aug. 2018
- Title: APISan: Sanitizing API Usages through Semantic Cross-checking
– Presented at USENIX Security 2016, Austin, TX Aug. 2016

Domestic

- Title: The Singularity is Here: When AI Meets Cyber Security
– Seminar at Tech-Share by Samsung Electronics, Hyundai Motor, and Naver, Seoul, Korea Feb. 2026
- Title: The Dark and Bright Sides of AI in Cybersecurity
– Seminar at Samsung Electronics DS, Seoul, Korea Jan. 2026
– Seminar at AI Security Technology Seminar, Seoul, Korea Dec. 2025
– Seminar at KAIST AI Graduate School Colloquium Oct. 2025
– Seminar at KISA, Seoul, Korea Oct. 2025
– Seminar at KISIA, Seoul, Korea Oct. 2025
– Seminar at KAIST-Cyber Security Policy Program Sep. 2025
– Seminar at Samsung SDS, Seoul, Korea Sep. 2025
– Seminar at National Intelligence Service, Seoul, Korea Sep. 2025
- Title: AI that Changed the Cybersecurity Talent and Education
– **Keynote** at Cybersecurity Talent Development Symposium, Seoul, Korea Dec. 2025
– Seminar at Cyber Security Policy Forum, Seoul, Korea Dec. 2025
- Title: The Future of Security Transformed by AI
– Seminar at ACDC (AI Cyber Defense Contest), Seoul, Korea Dec. 2025
- Title: AI that Changed the Battlefield
– **Keynote** at WhiteHat 2025, Seoul, Korea Nov. 2025
- Title: Atlantis: an Autonomous LLM-Powered Bug Finding and Fixing System
– Seminar at Cyber Security Society, Seoul, Korea Nov. 2025
– Seminar at Cyber Summit Korea 2025, Seoul, Korea Sep. 2025
– Seminar at Tech-Share by Samsung Electronics, Hyundai Motor, and Naver, Seoul, Korea Aug. 2025
– Seminar at ENKI WhiteHat, Seoul, Korea Sep. 2025
– Seminar at LIGNex1, Seoul, Korea Sep. 2025
– Seminar at KCSCON, Seoul, Korea Sep. 2025
- Title: AIxCC: The Future of Cybersecurity with LLMs
– Seminar at ETRI, Daejeon, Korea Nov. 2024
– Seminar at Moon Sool Graduate School of Future Strategy, Daejeon, Korea Nov. 2024
– Seminar at Future Security Tech Forum, Jeju, Korea Nov. 2024
– Seminar at National Security Research Institute (NSRI), Daejeon, Korea Sep. 2024
- Title: 2024 Security Strategy: Polarization
– Seminar at Defense Counterintelligence Command, Gwacheon, Korea May. 2024
- Title: Building Automated Hacking Systems
– Seminar at POSTECH, Pohang, Korea Nov. 2023

Title: Trends in Security Vulnerabilities of Low Earth Orbit Satellites	
– Presented at ETRI, Daejeon, Korea	Aug. 2023
Title: Academic Research from Offensive Research	
– Presented at Samsung, Seoul, Korea	Aug. 2023
Title: Human-friendly binary analysis	
– Presented at ETRI, Daejeon, Korea	Nov. 2023
– Presented at Korea Computer Congress (KCC), Seoul, Korea	Jun. 2023
Title: Exploit in the wild	
– Presented at ETRI, Daejeon	Jun. 2023
Title: Hacking 101	
– Presented at WISC, Seoul	Sep. 2022
Title: Attack and Defenses for Heap Vulnerabilities in 2022	
– Seminar at ETRI, Daejeon	Apr. 2022
Title: Comparative Analysis of Baseband Software and Cellular Specifications for Finding Vulnerabilities	
– Seminar at UNIST, Ulsan	May. 2023
– Seminar at Security@KAIST, Online	Jun. 2022
– Seminar at Cyber Operations Command, Seoul	Jun. 2022
Title: Scalable and Automatic Vulnerability Discovery Beyond Random Testing	
Seminar at Seoul National University, Seoul, Korea, Mar. 2019	
Title: Memory Allocator Security	
– Presented at Best of Best (BoB), Seoul	Feb. 2023
– Presented at Computer System Society Conference (CSC), Pyeongchang	Feb. 2023
– Seminar at UNIST, Online	May. 2022
– Seminar at Yonsei university, Online	Apr. 2022
– Seminar at Sungkyunkwan university, Online	Apr. 2022
– Seminar at ETRI, Daejeon	Jan. 2022
– Seminar at National Security Research Institute (NSRI), Daejeon	Dec. 2021
– Seminar at Securty@KAIST, Online	Nov. 2021
– Seminar at KAIST GSIS, Online	Nov. 2021
Title: Browser Security: Hacking & Research	
– Presented at Open Theori Research Seminar #6, Online	Dec. 2021
– Seminar at Hanyang University, Online	Nov. 2021
– Presented at KR Becks Meetup #1 by LINE, Online	Aug. 2021
– Seminar at Security@KAIST, Online	Jun. 2021

Grants

To summarize, 2.4B KRW is awarded, and my portion is 1.96B KRW.

- | | |
|---|---------------|
| 19. Intelligent multi-agent system for variant vulnerability detection in large-scale software | 26.03 – 27.02 |
| Agency/Company: NRF | |
| Money: 142M KRW | |
| Role: PI | |
| 18. Design and implementation of novel fuzzing system for USB driver security | 25.08 – 26.05 |
| Agency/Company: Samsung Electronics | |
| Money: 82.5M KRW | |
| Role: PI | |

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| 17. SAT-HACK: Satellite cyber kill chain verification research | 25.07 – 27.02 |
| Agency/Company: LIGNex1 | |
| Money: 275M KRW | |
| Role: PI | |
| 16. Software safety research based on automated vulnerability pattern generation | 25.04 – 25.10 |
| Agency/Company: NSRI | |
| Money: 60M KRW | |
| Role: PI | |
| 15. AIxCC: Automating real-world vulnerability discovery and remediation | 24.09 – 26.02 |
| Agency/Company: NYU | |
| Money: 99.9M KRW | |
| Role: PI | |
| 14. Research on building an open source kernel security model | 24.04 – 24.10 |
| Agency/Company: NSRI | |
| Money: 54.5M KRW | |
| Role: PI | |
| 13. Revisiting IoT threat models for smart cities and developing a vulnerability analysis system based on these models | 24.01 – 25.12 |
| Agency/Company: IITP | |
| Money: 400M KRW | |
| Role: PI | |
| 12. Building a system to assist variant analysis for browsers | 23.06 – 24.05 |
| Agency/Company: NRF | |
| Money: 65M KRW | |
| Role: PI | |
| 11. Generating a security model based on JavaScript intermediate language | 23.04 – 23.10 |
| Agency/Company: NSRI | |
| Money: 54.5M KRW | |
| Role: PI | |
| 10. Verifying security threats in open-source operating systems | 23.04 – 23.10 |
| Agency/Company: NSRI | |
| Money: 54.5M KRW | |
| Role: PI | |
| 9. Browser fuzzing with formal verification for cross architectures | 22.09 – 23.09 |
| Agency/Company: NRF | |
| Money: 110M KRW | |
| Role: PI | |
| 8. Building test suites for validating vulnerability detection | 22.08 – 22.11 |
| Agency/Company: ETRI | |
| Money: 27.3M KRW | |
| Role: PI | |
| 7. Generating a security model based on JavaScript security analysis | 22.04 – 22.10 |
| Agency/Company: NSRI | |

- Money: 54.5M KRW
Role: PI
6. **Developing techniques for collection and integrated analysis of automotive systems through event-based experimental systems** 22.04 – 23.12
Agency/Company: Dankook university
Money: 300M $\times$ 0.5 KRW
Role: Co-PI with Prof. Yujip Won
 5. **6G security** 21.08 – 23.09
Agency/Company: Samsung Electronics
Money: 200M $\times$ 0.2 KRW
Role: Co-PI with Prof. Yongdae Kim
 4. **DRAM security** 21.07 – 24.06
Agency/Company: Samsung Electronics
Money: 180M $\times$ 0.2 KRW
Role: Co-PI with Prof. Yongdae Kim
 3. **Systematic and precise transformation of the Qualcomm Hexagon architecture into intermediate representations for binary analysis** 21.06 – 22.05
Agency/Company: NRF
Money: 46.7M KRW
Role: PI
 2. **Automatically generating a security model for discovering web browser vulnerabilities** 21.04 – 21.10
Agency/Company: NSRI
Money: 54.5M KRW
Role: PI
 1. **Developing a scalable cyber reasoning system (Start-up)** 21.02 – 24.12
Agency/Company: KAIST
Money: 150M KRW
Role: PI

Advising and Mentoring

Ph.D. Students

- Youngmin Choi (Co-advising with Prof. Kyoungsoo Park)
- Haein Lee
- Sujin Han
- Junyoung Park
- Minwoo Baek
- Eunkyu Lee
- Donghyeon Kim
- Dongjun Lee

Master Students

- Kyeongmin Kim

- Juhyun Song
- Minwoo Jeong
- Gunhee Ahn

Alumni (Ph.D.)

- HyungSeok Han (Co-advised with Yongdae Kim), Postdoc at Georgia Tech Ph.D. in 2023

Alumni (M.S.)

- Dong-uk Kim, 78ResearchLab M.S. in 2025
- Dongok Kim, ENKI WhiteHat M.S. in 2025
- Hyeon Heo, ENKI WhiteHat M.S. in 2025
- Seunggi Min, Financial Security Institute M.S. in 2025
- Wonyoung Kim, Samsung M.S. in 2025
- Taisic Yun (Co-advised with Yongdae Kim), Theori M.S. in 2024
- Wonyoung Jung, 78ResearchLab M.S. in 2024
- Hyunsik Jeong (Co-advised with Yongdae Kim), S2W M.S. in 2021